

Quiz - Lecture 05

Close your laptops and put away all electronics. No electronics or notes are allowed during the duration of this quiz.

Question 1

What does the following code do?

```
Python
jupyterturtle.make_turtle()
```

- A. Imports the turtle module
- B. Creates a window for drawing
- C. Creates a turtle object
- D. Moves the turtle forward

Question 2

Which statement best describes what `jupyterturtle.left(90)` does?

- A. Moves the turtle 90 units left
- B. Rotates the turtle 90 degrees counterclockwise
- C. Rotates the turtle 90 degrees clockwise
- D. Turns the turtle to face west

Question 3

Why is this function definition useful?

```
Python
def square():
    for i in range(4):
        forward(100)
        left(90)
```

- A. It reduces indentation errors
- B. It avoids using loops
- C. It makes the turtle move faster
- D. It allows code reuse

Question 4

What shape does this code draw?

Python

```
for i in range(4):  
    forward(50)  
    left(90)
```

- A. Triangle
- B. Rectangle
- C. Square
- D. Circle

Question 5

If the turtle starts facing right (East), what direction is it facing after running this code?

Python

```
left(90)  
left(90)
```

- A. Up
- B. Left
- C. Down
- D. Right

Question 6

How many times does `forward(100)` execute?

Python

```
for i in range(1):  
    forward(100)
```

- A. 0
- B. 1
- C. 100
- D. Infinite

Question 7

What effect does changing `length` have?

Python

```
def line(length):  
    forward(length)  
  
line(200)
```

- A. Changes the turtle's speed
- B. Changes the turning angle
- C. Changes how far the turtle moves
- D. Changes pen thickness

Question 8

If the turtle starts by facing up (North), what direction is it facing after calling `turn()` once?

Python

```
def turn():  
    right(90)
```

- A. Up
- B. Down
- C. Left
- D. Right

Question 9

What is a function?

- A. A loop that repeats code
- B. A variable that stores numbers
- C. A drawing made by the turtle
- D. A block of code that performs a task

Question 10

Why do we use functions?

- A. To reuse code and keep programs organized
- B. To make the turtle move faster
- C. To avoid writing Python code
- D. To remove loops

Bonus Question 1

What shape will this draw?

Python

```
def square():  
    for i in range(3):  
        forward(100)  
        left(90)
```

- A. Pentagon
- B. Triangle
- C. Square
- D. Incomplete Square

Bonus Question 2

What shape will this draw?

Python

```
def square(length):  
    for i in range(4):  
        forward(length)  
        left(90)
```

```
square(50)  
square(100)
```

- A. One square with side length 150
- B. Two squares of the same size
- C. One rectangle
- D. Two squares of different sizes

Bonus Question 3

Write code that creates a turtle, then uses a for loop to draw 36 squares at size 100, rotating the turtle 10 degrees left after each square. You do not have to define the `square()` function.

Assume `jupyterturtle` is already imported.